Chemical Engineering Science 338 (2027) 125093

Contents lists available at ScienceDirect

Chemical Engineering Science

journal homepage: www.elsevier.com/locate/ces

Tuning of physicochemical and photocatalytic properties of dual-phase TiO_2: Rare earth mono-doping versus co-doping☆

Tetiana Khalyavka [a,b,*], Regina Abdurrahmanoglu [a], Nataliya Shcherban [c], Mykhailo Chaika [d], Tetiana Gavrilko [d,e], Oscar Cabezuelo [f,g], Oleg Tkachenko [f,h], Elif Coşkun [a], Pavel Yaremov [c], Tetyana M. Budnyak [f,g,h], Jean-Claude Grivel [a]

[a] Department of Energy Conversion and Storage, Technical University of Denmark, Kongens Lyngby, 2800 Denmark
[b] Institute for Sorption and Problems of Endoecology, NASU, 13 Oleg Mudrak Str., 03164 Kyiv, Ukraine
[c] L.V. Pisarzhevskii Institute of Physical Chemistry, NASU, 31 Nauky Ave., Kyiv 03028, Ukraine
[d] Institute of Low Temperature and Structure Research, Polish Academy of Sciences, ul. Okólna 2, 50-422 Wrocław, Poland
[e] Institute of Physics NASU, 46 Nauky Av., 02028 Kyiv, Ukraine
[f] Department of Earth Sciences, Uppsala University, Villavägen 16, 752 36 Uppsala, Sweden
[g] Wallenberg Initiative Materials Science for Sustainability, Department of Earth Sciences, Uppsala University, Villavägen 16, 752 36 Uppsala, Sweden
[h] Division of Nanotechnology and Functional Materials, Department of Materials Science and Engineering, the Ångström Laboratory, Uppsala University, Uppsala 751 03, Sweden

ARTICLE INFO

Keywords:
Anatase-brookite dual-phasetitania
Co-doping
Doping
Rare earth elements
Hydrothermal method
Photocatalysis

ABSTRACT

Modification of TiO_2 with rare-earth elements is an effective approach to improve photocatalytic efficiency and extend photoactivity into the visible region. Dual-phase anatase-brookite TiO_2 mono-doped with Sm, Gd, Tb, Tm, or Yb, and co-doped with SmGd, SmTb, SmTm, or SmYb, were prepared by a template-free one-step hydrothermal method, and the effects of mono- versus co-doping on the morphological, structural, textural, optical, and photocatalytic properties of the materials were systematically compared. X-ray diffraction, Raman, Fourier-transform infrared, and high-resolution transmission electron microscopy confirmed that the dual-phase anatase-brookite structure is preserved after doping. The formation of mesoporous materials with developed porosity was proven by nitrogen physisorption. It was established that mono-doping favours the formation of powders with increased surface area, while co-doping supports the formation of larger pores. Ultraviolet–visible diffuse reflectance spectroscopy and photoluminescence spectroscopy demonstrated enhanced visible-light absorption and suppressed charge recombination for the doped powders, with stronger effect for co-doped materials due to the synergistic contribution of coupled dopants' energy levels. The photocatalytic performance was evaluated in the degradation of a commercial diclofenac drug under UV and visible-light illumination. In the tests, the most active catalyst was TiO_2 co-doped with Sm and Gd, which reached the drug removal rates of 20.4 % and 79.9 % under visible (>450 nm) and ultraviolet light, respectively. These results provide new insight into the structure–property–activity relationships of mono- and co-doped REE anatase-brookite TiO_2 composition and will aid the optimal design and fabrication of photocatalysts for pharmaceutical waste decomposition in aquatic environments.

1. Introduction

Nowadays, rare earth elements (REE) are widely used across various fields, and their application tends to grow (Balaram, 2019; Liu et al., 2023). Due to their unique magnetic, phosphorescent, and catalytic properties, these elements are applied in catalysis, electrocatalysis, electronics, transportation, and medicine (Balaram, 2019; Xu et al., 2024; Wang et al., 2023). Much attention is paid to designing photocatalysts (Mikolajczyk et al., 2024; Liu et al., 2018; Wang et al., 2015) with REE that result in materials with improved properties (Yuqi et al., 2024; Alviz-Meza et al., 2024; Bellardita et al., 2011).

Photocatalysis as a sustainable green technology deserves special

☆ This article is part of a special issue entitled: 'Emerging Trends in Semiconductor Photochemistry' published in Chemical Engineering Science.

* Corresponding author.
E-mail addresses: tekha@dtu.dk, takhalyavka@ukr.net (T. Khalyavka).

https://doi.org/10.1016/j.ces.2026.125093
Received 11 February 2026; Received in revised form 7 August 2026; Accepted 4 September 2026
Available online 7 September 2026
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attention. This method allows one to destroy different types of water contaminants (Liu et al., 2018; Wang et al., 2026; Tran et al., 2017). The most extensively studied photocatalyst is titanium dioxide (TiO_2), which unfortunately can only be excited by ultraviolet (UV) rather than visible (Vis) light. To overcome this limitation, doping and co-doping with different dopants, including REE, are applied (Mikolajczyk et al., 2024; Kramar et al., 2024; Reszczyńska et al., 2014).

Incorporating REE as dopants in TiO_2 offers several benefits. They can modify the electronic structure of TiO_2, introducing new energy levels within the band gap that enhance charge separation and extend light absorption into the visible range. Moreover, *f*-orbitals of REE can facilitate electron transfer and influence surface chemistry, thereby improving photocatalytic efficiency (Di Paola et al., 2010; Wu et al., 2019; Xu et al., 2002; Kumari et al., 2020). In particular, Sm, which has been proven to be an effective dopant for improving the TiO_2 photoresponse, increases the efficiency of photogenerated electron–hole pair separation and improves charge transfer (Yuqi et al., 2024; Di Paola et al., 2010). It can introduce localized energy states, enhance solar absorption in the visible range, and create luminescent centers, enabling absorption edge downshifting processes beneficial for solar energy applications (Wang et al., 2025; Suresh and Jayasankar, 2018; Guetni et al., 2020). Gd as a dopant induces lattice distortion in TiO_2 and generates more oxygen vacancies, which act as electron traps (Wu et al., 2019; Xu et al., 2002). Fewer reports are devoted to REE co-doping of TiO_2, and most of them focus on the co-doping effect on the physicochemical characteristics of TiO_2 rather than its photocatalytic activity (Trabelsi et al., 2020; Katta et al., 2023).

Both single REE doping and co-doping also contribute to one of the important stages in photocatalysis: substrate adsorption. The increase in the surface area of REE-doped and co-doped photocatalyst increases the number of adsorption and photocatalytic active centres (Trabelsi et al., 2020; Katta et al., 2023; Shi et al., 2011; Luo et al., 2017). It has been confirmed (Bellardita et al., 2011; Xu et al., 2002) that REE with their partially filled 4*f* orbitals and vacant 5*d* orbitals can easily combine with functional groups of reactant molecules (substrates), contributing to their better adsorption.

The bandgap tunning of TiO_2 with REE doping or co-doping has become a significant way to improve its photocatalytic performance (Mazierski et al., 2019; Borrego-Pérez et al., 2020; De Los Santos et al., 2015). The large proportion of the reports confirmed a slight red shift in UV–Vis diffuse reflectance spectra for REE doped or co-doped TiO_2. Thus, the modification with Tm (Mazierski et al., 2019); Gd, Nd, La, Pr, Er, Ce and Sm (Xu et al., 2002), Sm, Y, Ce, La (Yuqi et al., 2024), Tb (Kumari et al., 2020); Er, Nd (Katta et al., 2023); Ho, Yb (Nain et al., 2025), Y/Eu (Wang et al., 2015), La/Eu (Shi et al., 2011); Er/Yb (Wang et al., 2017); Ho/Yb (Nain et al., 2025; Matakgane et al., 2024), Ce (Kramar et al., 2024); Nd/Yb (Khan et al., 2024), Ce/Nd (Nain et al., 2024) favors TiO_2 band gap narrowing. At the same time, doping with Er (Rao et al., 2019); Nd (Labreche et al., 2018; Umadevi et al., 2022), La (Khalyavka et al., 2021), and co-doping with Er/Nd (Katta et al., 2023) and Sm/Eu (Zhang et al., 2019) led to an increase of band gap value. REE in co-doped TiO_2 can also participate in the up-conversion mechanism, transforming near-infrared or visible light into visible or UV light, respectively, which expands light harvesting and overcomes the wide bandgap limitations of TiO_2, enabling it to be activated by a larger portion of the solar spectrum (Patra et al., 2003; Madan et al., 2022).

Most of the studies have been carried out with anatase (Mikolajczyk et al., 2024; Reszczyńska et al., 2014; Trabelsi et al., 2020; Coromelci et al., 2022) or anatase–rutile TiO_2 (Katta et al., 2023; Shi et al., 2012) doped or co-doped with REE. The anatase-brookite composition remains less studied despite its advantages. As reported in Nachit et al. (2022); anatase-brookite heterojunctions provide more efficient charge separation than the widely studied anatase–rutile system, owing to electron transfer from the conduction band of brookite to that of anatase, arising from their different potentials, which suppresses electron-hole recombination. Brookite alone possesses a favourable electron trap depth that prolongs the lifetime of photogenerated charge carriers, while the distinctive coordination of its surface oxygen atoms is thought to further contribute to catalytic activity (Vequizo et al., 2017). It was shown that this composition exhibited even higher activity than anatase–rutile TiO_2 (Nachit et al., 2022; Morais et al., 2026). Our previous results (Khaliavka et al., 2025; Khalyavka et al., 2025) confirmed that modification of mixed anatase-brookite TiO_2 composition with REE led to an increase of photocatalytic activity compared to unmodified TiO_2. To the best of our knowledge, almost no reports on the photocatalytic activity of such a composition co-doped with REE have been published.

In the current study, dual-phase anatase-brookite TiO_2 co-doped with SmTm, SmGd, SmTb, SmYb and mono-doped with Sm, Gd, Tb, Tm, Yb have been prepared. The chosen dopants have different ionic radii: Sm^{3+} (0.0958 nm) > Gd^{3+} (0.0938 nm) > Tb^{3+} (0.092 nm) > Tm^{3+} (0.087 nm) > Yb^{3+} (0.086 nm), which enables examination the relationship between the ionic size mismatch relative to Ti^{4+} (0.065 nm) and structural distortion. Besides, the selected REE represent different types of electronic 4*f* configurations, which allow us to examine their interaction with the same anatase-brookite TiO_2 host.

The main visible-light-sensitizing dopant Sm^{3+} ($4f^5$) is a well-known luminescent center in TiO_2, which exhibits host-sensitized emission that is more efficient than direct excitation of the Sm^{3+} ion (Chang et al., 2017). It makes Sm suitable for extending visible light absorption and studying the energy transfer process from host to dopant. Tb^{3+} ($4f^8$) is also recognized as an efficient luminescent center with characteristic green emission (Hu et al., 2020), whose behavior in the TiO_2 host needs to be examined. When paired with Sm, it can enhance Sm^{3+} luminescence via energy transfer from Tb^{3+} (sensitizer) to Sm^{3+} (activator) (Di et al., 2026). Gd^{3+} ($4f^7$) with a half-filled shell, has its lowest excited state far in the UV region and does not show visible emission, making it suitable for dopant-to-host energy transfer processes (Zhang et al., 2026). Gd can act as a sensitizer transferring energy to Sm and significantly boosting emission output (Li et al., 2025). Tm^{3+} ($4f^{12}$) and Yb^{3+} ($4f^{13}$) have electronic configurations close to the fully filled $4f^{14}$. They were included in the list of dopants for comparison with lighter lanthanides having larger radii and different optical behaviour (Conrad et al., 2025). Both these dopants typically emit in the near-infrared and do not have emission in the range investigated in this work. Coupling Sm with each dopant, such as Gd, Tb, Tm, Yb allows us to test potential synergistic effects.

In the present work, the effect of REE doping and co-doping on morphological, structural, and optical characteristics of anatase-brookite TiO_2 has been studied. The photocatalytic properties of the prepared materials were assessed in the reaction of diclofenac (DCF) degradation under UV and visible light (Vis). DCF is a non-steroidal anti-inflammatory drug and one of the most commonly used medicines around the world. It was chosen because it is often found in aquatic environments and can be effectively degraded using photocatalytic technology. Contrary to other reports on photocatalytic degradation of DCF, where the model solutions were prepared from diclofenac sodium salt for photodegradation (Fergani et al., 2025; Xue et al., 2025; Lee et al., 2021; Thiruppathi et al., 2017), we applied a commercial medicine.

Thus, in this work, for the first time, we studied the photocatalytic activity of REE- doped and co-doped anatase-brookite TiO_2 in the DCF photodegradation reaction, along with morphological, structural, and optical characterization of the prepared materials. This research contributes to advancing knowledge in the field, providing new insights into the improvement of photocatalytic efficiency of TiO_2 for environmental applications.

2. Experimental

2.1. Materials and synthesis of photocatalysts

Details are provided in the Supplementary Information Section 1.

The synthesized undoped anatase-brookite TiO_2 composition was labelled as T, mono-doped materials as T_{Sm}, T_{Gd}, T_{Tb}, T_{Yb}, and T_{Tm}, co-doped samples as T_{SmGd}, T_{SmTb}, T_{SmTm}, and T_{SmYb}.

2.2. Characterization of synthesized powders

The detailed part was in the Supplementary Information Section 2.

2.3. Studies of photocatalytic properties

For the photocatalytic activity tests, Diclofenac (Darnytsia, Ukraine) was used as the target pharmaceutical compound. It contains diclofenac (DCF, $C_{14}H_{11}Cl_2NO_2$, 25 mg/mL) as the main active substance, together with excipients such as mannitol ($C_6H_{14}O_6$), sodium metabisulfite ($Na_2S_2O_5$), benzyl alcohol (C_7H_8O), propylene glycol ($C_3H_8O_2$), sodium hydroxide (NaOH) and water for injections. The commercial DCF was diluted with distilled water to reach a concentration (C_0) of DCF 5.5 × 10^{-5} mol/L (pH ~ 7).

The photocatalyst dosage was 1 mg/ml. A VWR® UV-3100PC UV/Vis spectrophotometer was used to determine the DCF concentration by measuring the absorbance of the solutions at 275 nm. In this wavelength region DCF exhibits a well-defined absorption band whereas mannitol and propylene glycol have no significant UV–Vis chromophore and are not expected to interfere at this wavelength, while sodium metabisulfite and benzyl alcohol could, in principle, contribute a minor background absorbance.

Before the photocatalytic experiments the solutions were kept in the dark and stirred for 2 h to reach the adsorption/desorption equilibrium. The adsorption removal (R_A, %) was evaluated using the equation: $R_A = (C_{in} - C_0)/C_{in} \times 100$, where C_{in} is the initial concentration of the DCF, C_0 is the concentration of the DCF solution after adsorption.

The DCF photodegradation was then carried out in the presence of photocatalysts under visible-light (Vis) (a LED lamp, $\lambda > 450$ nm) and UV (a UVP 3UV Lamp, 8 W power, $\lambda_{max} = 365$ nm) illumination for 3 h

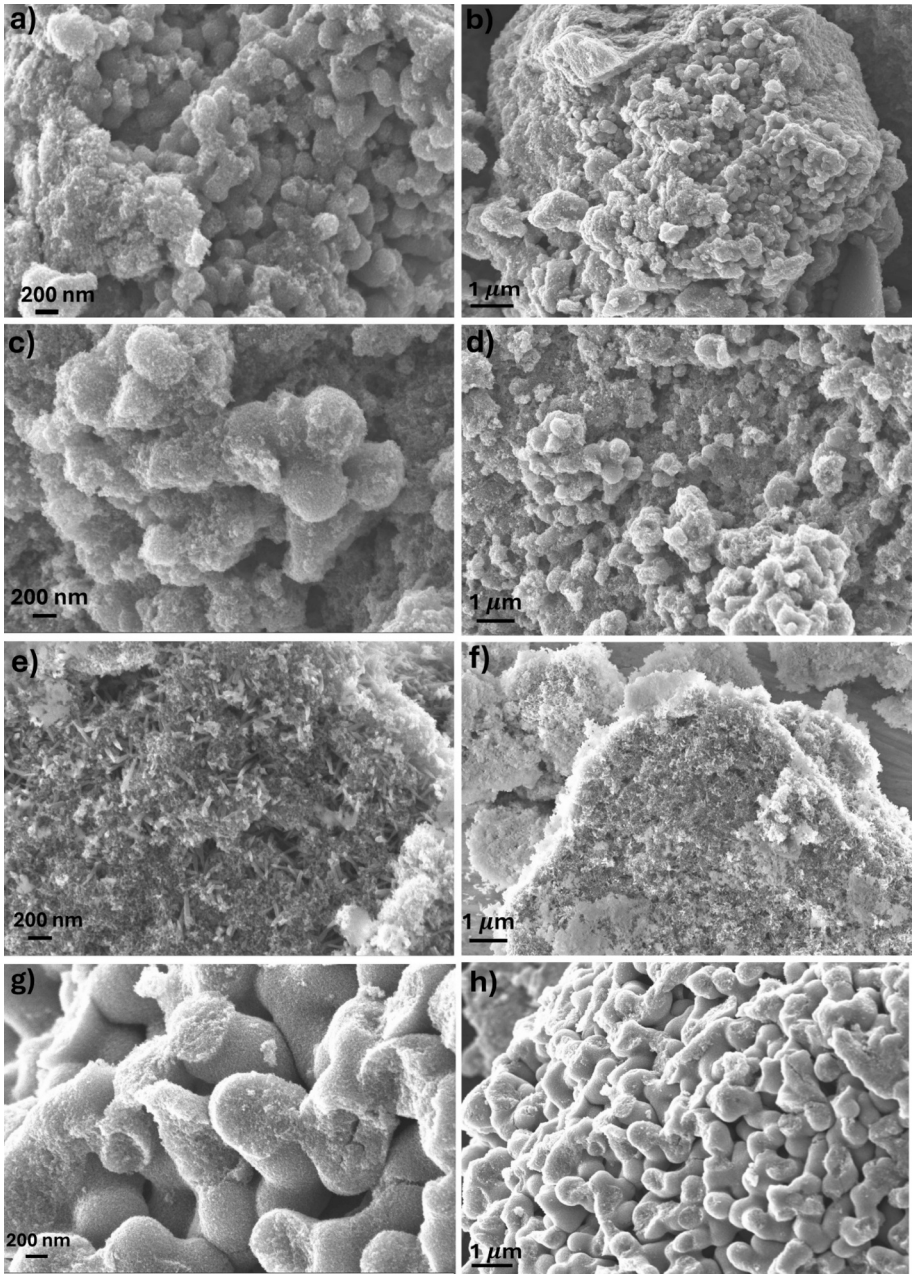


Fig. 1. SEM images for T (a, b), T_{Sm} (c, d), T_{Gd} (e, f), and T_{SmGd} (g, h).

at room temperature in air. Control experiments were also performed to evaluate the photolysis of DCF under UV and Vis illumination in the absence of a photocatalyst.

The percentage of DCF removal (R_{Ph}, %) under illumination was derived from the difference between C_0 and DCF concentration after 3 h of illumination in the presence of photocatalyst (C_{ph}): $R_{Ph} = (C_0 - C_{ph})/C_0 \times 100$. The DCF removal value (R, %) was calculated using the following equation: $R = R_A + R_{Ph}$.

The photocatalytic reaction follows the pseudo-first-order kinetic model: $\ln(C_0/C_{ph}) = -kt$, where k is the pseudo-first-order reaction rate constant, and t is reaction time. The photocatalytic rate constant was calculated from the slope of the linear plot of $\ln(C_0/C_{ph})$ versus time. All experiments were performed in triplicate, and the results are reported as mean values ± standard deviation.

3. Results and Discussion

3.1. Morphological and elemental characterization of the powders

To investigate the morphology of the REE-doped powders, SEM characterization was performed. As can be seen from Fig. 1, the prepared materials consist of agglomerates with different shapes. The agglomerates of undoped TiO_2 contain roundish particles with a smooth surface (Fig. 1, a, b), whereas the powder doped with Sm consists of loosely structured roundish formations (Fig. 1, c, d). The sample doped with Gd comprises elongated, needle-like particles (Fig. 1, e, f). Co-doping of TiO_2 with Sm and Gd led to the formation of loose, filamentous, sinuous particles containing voids (Fig. 1, g, h), suggesting a well-developed porous structure that may be favourable for substrate adsorption.

Despite the different shapes of agglomerates, the TEM images are similar and show diamond-shaped crystallites with an average size around 10 nm (Fig. 2, Fig. S1-S3).

EDS was applied to estimate the composition and verify the presence of REE in the powders of doped TiO_2. The studies revealed that the materials contain titanium, oxygen, and carbon (this might have come from the carbon tape used during sample preparation). The presence of REE dopants in REE mono- and co-doped TiO_2 was confirmed (Fig. S4).

XPS method was also used to verify the elemental content and the charge state of atoms in the prepared materials. In the survey scans (Fig. S5-S10), peaks corresponding to all expected elements were detected, which was in accordance with the EDS analysis (Fig. S4).

The samples displayed characteristic doublet peaks at around 459 eV (Ti $2p_{3/2}$) and 465 eV (Ti $2p_{1/2}$) (Fig. S5-S10). The spin–orbit splitting value of 5.7 eV between these peaks confirmed the presence of TiO_2, and the observed binding energies are characteristic of titanium in the + 4-oxidation state (Erdem et al., 2001; Wagner et al., 2003). Deconvolution of the asymmetric O 1 *s* signal revealed three sub-peaks, indicating the presence of different oxygen species on the material surface. The first peak, centred at around 530.6 eV, corresponds to Ti–O bonds, while the second, at 531.8 eV, is assigned to the surface hydroxyl groups (Ti–OH) (McCafferty and Wightman, 1998) and the third peak around 532 eV is attribute to the oxygen-containing surface species (Frankcombe and Liu, 2023).

Two peaks were obtained after the deconvolution of Gd 4*d* signal in the XPS spectra of T_{Gd} and T_{SmGd} (Fig. S6-S7). The first peak at around 143.5 eV and the second one at 148.8 originate from Gd $4d_{5/2}$ and Gd $4d_{3/2}$ and correspond to the oxidation state + 3 of Gd (Sarkar et al., 2024; Saha et al., 2017). The main valence state of Sm in the mono- and co-doped samples is + 3, while a small fraction of divalent samarium Sm^{2+} ($3d_{5/2}$) was observed at about 1073.9 eV for T_{Sm} and T_{SmGd} (Brunckova et al., 2018). The Sm 4*d* region exhibits peaks at 137.45 eV and 132.85 eV, which are assigned to the Sm $4d_{5/2}$ and Sm $4d_{3/2}$ spin–orbit components, respectively, further confirming the presence of Sm^{3+} (Fig. S8-S10). Yb in T_{Yb} and T_{SmYb} exists as Yb^{3+} with peaks at around 186 eV and 193 eV, which originate from the $4d_{5/2}$ and $4d_{3/2}$ components, respectively (Fig. S8) (Loyal et al., 2023). The deconvoluted spectra in the 4*d* region for Tm showed a $4d_{3/2}$ peak at a binding energy of around 177 eV, accompanied by satellite peaks at higher energy, indicating the presence of Tm^{3+} oxidation state (Fig. S9) (Kabongo et al., 2017). The Tb 4*d* region shows spin–orbit split peaks at binding energies of 157.25 eV and 151.1 eV, corresponding to the Tb $4d_{5/2}$ and Tb $4d_{3/2}$ components, respectively, and consistent with the + 3-oxidation state of Tb (Fig. S10) (Gnanasekar et al., 1994).

3.2. Structure characterization

The XRD, Raman, FTIR, and HR-TEM methods were applied to analyse the structure of the studied materials. All XRD patterns (Fig. 3, a) and Raman spectra (Fig. 3, b, Fig. S11) match well with the crystal phases of anatase (A) and brookite (B) (Tran et al., 2017).

The Rietveld refinement of the XRD patterns additionally confirmed the formation of these two crystalline phases of TiO_2 and the decrease of the unit cell volume and lattice parameters (Table S1). Since the dopants have larger ionic radii (0.0958 nm for Sm, 0.0938 nm for Gd, 0.092 nm for Tb, 0.087 nm for Tm, 0.086 nm for Yb) than Ti^{4+} (0.065 nm) they induce lattice distortion. As can be seen from Table S1, co-doping induces a larger degree of cell contraction.

Peaks associated with REE cannot be observed in the XRD patterns due to their low concentration in the samples (Fig. 3, a) (Mikolajczyk et al., 2024; Bashir et al., 2023). REE cannot be substitutionally introduced into the TiO_2 lattice; instead, they are more likely accommodated in interfacial (anatase-brookite junction) sites and/or segregated at the grain boundaries forming small clusters (Fig. S12). The average values of the crystallite size for the synthesized samples were calculated from XRD data (Table 1).

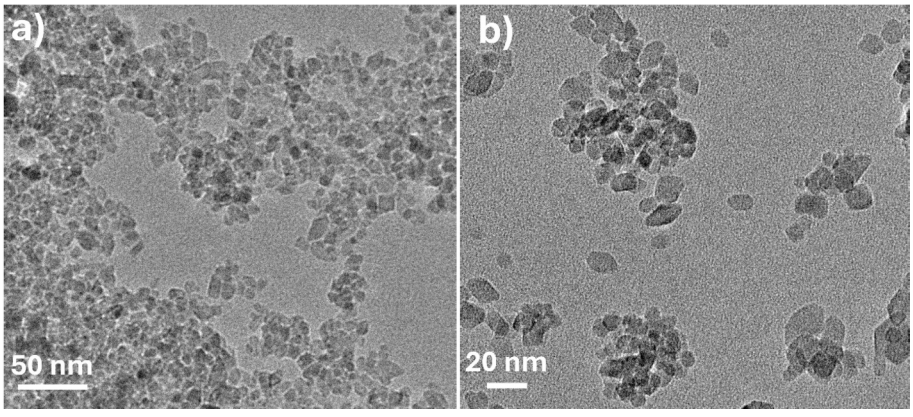


Fig. 2. TEM images of T_{Gd} (a), and T_{SmGd} (b).

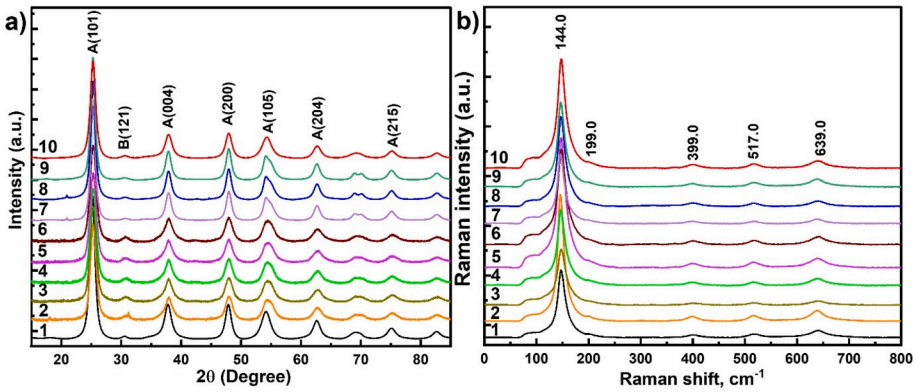


Fig. 3. XRD patterns (a) and Raman spectra (b) of the powders: T (1), T_{Sm} (2), T_{Yb} (3), T_{Tb} (4), T_{Gd} (5), T_{Tm} (6), T_{SmGd} (7), T_{SmTb} (8), T_{SmYb} (9), and T_{SmTm} (10).

Table 1
The average crystallite size (D_{Sch}*, nm), specific surface area (S_{BET}, m^2/g), total pore volume (V_{tot}, cm^3/g) and pore diameter (D_{pore}, nm) of the samples.

Parameter	Sample									
	T	T_{Sm}	T_{Gd}	T_{Tb}	T_{Tm}	T_{Yb}	T_{SmGd}	T_{SmTb}	T_{SmTm}	T_{SmYb}
D_{Sch} A (101)	9	8	7	7	7	7	10	10	8	10
D_{Sch} B** (121)	8	8	7	7	7	7	7	6	6	5
S_{BET}	165	185	190	190	185	195	160	160	165	150
V_{tot}	0.27	0.31	0.30	0.30	0.29	0.30	0.30	0.31	0.30	0.31
D_{pore}	5.9 ± 0.5	5.7 ± 0.5	5.8 ± 0.5	6.2 ± 0.6	5.6 ± 0.6	5.4 ± 0.5	6.7± 2.1	6.4± 1.5	6.3± 0.8	6.7± 1.8

* Values rounded to the nearest whole integer (half-integers rounded up) results; A – anatase, B – brookite. **The B crystallite size was calculated from a single, low-intensity (121) reflection and should be regarded as an approximate estimate.

Contrary to mono-doping, co-doping favoured a slight increase in the crystallite size of anatase. A similar increase in the crystallite size, together with lattice contraction, upon REE co-doping of TiO_2 has been observed previously in publications (Trabelsi et al., 2020; Matakgane et al., 2024; Khan et al., 2024).

Additionally, Raman spectroscopy, known for its high sensitivity to the phase composition, crystallite size, and stoichiometry of TiO_2 nanomaterials was applied to independently confirm both polymorphs detected by XRD with tetragonal anatase as the dominant phase in all powders (Fig. 3, b). Raman-active modes of anatase TiO_2 observed at around 147 (E_g), 199 (E_g), and 399 cm^{-1} (B_{1g}) are assigned to Ti–O bending, and those at 639 cm^{-1} (E_g) and 517 cm^{-1} ($A_{1g} + B_{1g}$) are referred to as Ti–O stretching vibrations of TiO_2 (Radha et al., 2022; Komaraiah et al., 2019). All samples also showed three additional weak Raman peaks at 245 (A_{1g}), 320 (B_{1g}), and 370 (B_{2g}) cm^{-1}, corresponding to the brookite phase (Tran et al., 2019) (Fig. S11). The low intensity of these peaks confirms the presence of brookite in small quantities. Apart from small deviations in relative intensities of brookite-related bands, there is no significant difference between the Raman spectra of the undoped and REE-doped samples. Due to the small dopant amounts, no peaks indicating the presence of REE were detected within the sensitivity limit of our Raman measurements.

The presence of anatase and brookite phases after doping was also confirmed by HR-TEM. The analysis of a T_{SmGd} TEM-image revealed the *d* spacings characteristic of anatase and brookite (Fig. S1-S3, S13).

Phase composition and molecular bonding in the prepared materials were further investigated with FTIR spectroscopy (Fig. 4).

Fig. 4 shows the FTIR absorption spectra of the undoped and REE-doped dual-phase TiO_2 in the mid-infrared (MIR, 380–4000 cm^{-1}, a) and far-infrared (FIR, 100–400 cm^{-1}, b) spectral ranges, covering stretching and bending vibrations of the Ti–O, Ti–O–Ti, and REE–O bonds, as well as fundamental vibrations of adsorbed water molecules and surface hydroxyl groups.

Based on the available literature data and the deconvolution of the

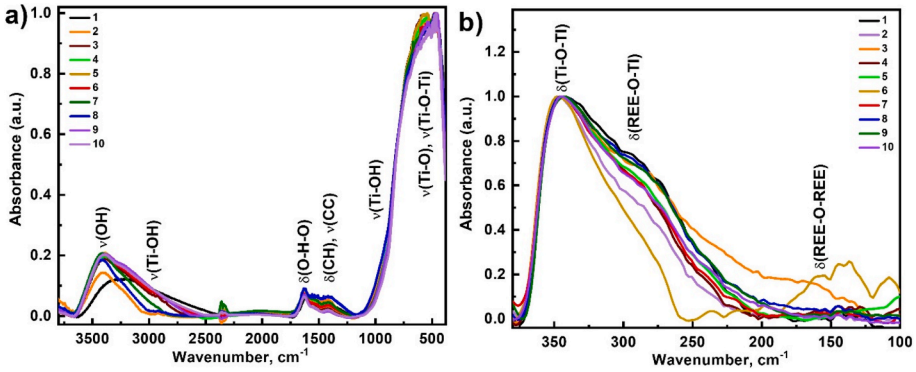


Fig. 4. FTIR absorption spectra (a) and absorption spectra in far-infrared spectral range (100–500 cm^{-1}) (b) of the samples: T (1), T_{Sm} (2), T_{Tm} (3), T_{Tb} (4), T_{Yb} (5), T_{SmTm} (6), T_{SmYb} (7), T_{SmTb} (8), T_{Gd} (9), and T_{SmGd} (10).

main absorption band of undoped TiO_2 with Gaussian functions (not shown here), the strongest absorption band component centred approximately at 540 cm^{-1}, is assigned to terminal ν(Ti–O) stretching vibrations, and a band with peak position at 470 cm^{-1} to bridging ν(Ti–O–Ti) stretching vibrations characteristic of the anatase phase. The weaker components at around 635 and 715 cm^{-1} are assigned to ν(Ti–O–Ti) and ν(Ti–O) modes in the brookite phase. Correspondingly, bending δ(Ti–O–Ti) and δ(Ti–O) modes are observed in the far-infrared region at approximately 342 and 281 cm^{-1} (Scarel et al., 2002). At high wavenumbers, a broad band centred at ~ 3200 cm^{-1} is assigned to ν(O–H) stretching of physically adsorbed water and surface Ti–O–H groups, while a δ(H–O–H) bending band of adsorbed moisture is present at 1640 cm^{-1}. The weaker bands with frequencies of 1417 and 1540 cm^{-1} originate from C–H bending and C–C stretching vibrations in precursor residues or carbonate species formed from the environment (Jackson and Parfitt, 1971).

FTIR absorption spectra of nanocrystalline REE compounds are characterized by prominent absorption bands in the region between 600 cm^{-1} and 1000 cm^{-1}, formed by overlapping broad absorption sub-bands originating from different ν(REE–O) and ν(O-REE–O) motions within the crystal lattice. For instance, Gd–O and O–Gd–O stretching vibrations appear at around 570 cm^{-1} and 440 cm^{-1} (Huong et al., 2015), the corresponding Sm–O modes are observed at roughly 548, 744, and 857 cm^{-1} (Jamnani et al., 2023), and La–O bending and stretching vibrations are located at around 530–620 cm^{-1} and 950–1100 cm^{-1}, respectively (Karthikeyan et al., 5 (2022)). δ(REE–O) bending vibrations are typically observed in the far-infrared region at about 180–340 cm^{-1}. Though the exact position of the corresponding IR absorption bands strongly depends on the material crystal phase, particle size, and morphology, the ν(REE–O), ν(O-REE–O) and δ(REE–O) main absorption bands are typically located close to the corresponding TiO_2 bands and strongly overlap with the dominant TiO_2 absorption. Therefore, it was hardly possible to qualitatively evaluate the doping effect on these vibrations, especially considering the small dopant concentration and its negligible effect on TiO_2 crystal lattice.

As seen from Fig. 4, the FTIR spectra of the REE-doped and undoped samples exhibit a generally similar pattern; however, the effect of doping on TiO_2 is evident indirectly, as a variation in intensity as well as small shifts in the Ti–O and Ti–O–Ti stretching and bending vibration bands. For instance, for undoped TiO_2, the stretching ν(Ti–O) band is observed at 541 cm^{-1}, while for T_{Gd} and T_{SmGd} samples it is shifted to 537 and 527 cm^{-1}, respectively. Additionally, the bridging ν(Ti–O–Ti) stretching vibration band for these samples is also shifted from 470 cm^{-1} for undoped TiO_2 to 464 cm^{-1} and 462 cm^{-1}, respectively. Similarly, in the far-infrared region, bending δ(Ti–O–Ti) mode is shifted from 342 cm^{-1} for undoped TiO_2 to 345 and 347 cm^{-1} for T_{Gd} and T_{SmGd} samples, respectively, owing to the contribution of δ(REE–O) modes. Thus, the observed IR band shifts and variation in their intensities indicate that the TiO_2 crystal lattice undergoes notable distortion upon REE doping in agreement with the XRD results.

The most significant changes in the FTIR spectra of REE-doped TiO_2 samples are detected in the OH/Ti–OH stretching bands at 3200–3600 cm^{-1}. The broad and asymmetric OH stretching absorption band can be deconvoluted into three Gaussian components corresponding to the vibrations of different types of hydrogen-bonded OH groups (Fig. S12). The first component, at 3400 cm^{-1}, is attributed to symmetric and asymmetric OH stretching vibrations of physisorbed water molecules bound by weak hydrogen bonds. The second band, at 3200 cm^{-1}, is assigned to water complexes strongly bound to the TiO_2 surface, and the third one, at about 3000 cm^{-1}, relates to OH stretching vibrations in Ti–OH/ REE-OH groups bonded to the nanoparticle surface by stronger hydrogen bonds. The changes observed in the OH stretching spectral range after REE doping, such as 1.5–2-fold total increase in the intensity of ν(OH) bands, imply an increase in the amount of adsorbed water and surface hydroxylation. Moreover, it is evident that the intensity of the Ti–OH / REE-OH band at 3000 cm^{-1} (Fig. S14), increases for T_{Gd} and T_{SmGd} samples, implying that they contain more oxygenated surface species, whereas T_{Sm}, T_{SmTb} and T_{SmYb} samples exhibit a slightly less intense Ti–OH band. Thus, REE doping-induced changes in surface hydroxyl coverage depend on the doping cation, which together with other factors, such as surface area, crystallite size, and morphology may be decisive for the photocatalytic performance of the REE-doped TiO_2 nanostructures.

3.3. Textural properties and porous structure of the materials

The results of BET analysis are in good agreement with SEM and XRD data. The values of surface area lie in the range of 185–195 m^2/g for mono-doped samples with smaller crystallite size, and 150–165 m^2/g for co-doped powders with larger crystallites; the values of total pore volume are in the range 0.29–0.31 cm^3/g and 0.30–0.31 cm^3/g, respectively (Table 1).

According to IUPAC classification all the isotherms of the samples belong to type IVa with an H2b type of hysteresis loop (Thommes et al., 2015; Lowell and Shields, 2013), which is characteristic of mesoporous materials (Fig. S15). The detailed analysis of isotherms revealed that undoped TiO_2 is characterized by a homogeneous mesoporous structure, due to the formation of particles of approximately the same size during the synthesis, whereas doped materials display a less homogeneous mesoporous structure. These results are consistent with the SEM analysis (Fig. 1).

The pore size distributions for the samples (Fig. 5) also confirmed their mesoporous nature with a negligible amount of micropores (~0.005 and 0.01 cm^3/g for mono- and co-doped materials, respectively).

It can be seen from Fig. 5 that the mono-doped samples have a pore size distribution similar to that of undoped TiO_2 (5.4–6.2 ± 0.5 nm). The formation of the mesoporous structure changed substantially during the co-doping process. In this case, similarly less homogeneous mesoporous structures were formed but with a larger average pore size: up to ~ 6.7 ± 2.1 nm for T_{SmGd}, ~6.4 ± 1.5 nm for T_{SmTb}, ~6.3 ± 0.8 nm for T_{SmTm}, and ~ 6.7 ± 1.8 nm for T_{SmYb} as compared to mono-doped materials (Table 1). Such changes in particle morphology and specific surface area may contribute to the improved photocatalytic activity of REE-doped TiO_2.

3.4. Optical properties of materials

The UV–Vis diffuse reflectance spectra were used to analyse the optical characteristics of both mono- and co-doped samples in comparison with undoped TiO_2, as displayed in Fig. 6.

The absorption edges of all the samples were close to ~400 nm. Also, trivalent REE ions absorb in the UV spectral range (125–400 nm), where they show strong broad absorption bands arising from the 4*f*-5*d* electron transitions. This absorption generally increases from Ce to Gd and drops in Tb (Sharma et al., 2020). However, for the materials under study, none of these bands can be analysed since they are completely overlaid with strong intrinsic absorption of the TiO_2 host.

The light absorption intensity between 400 and 550 nm increased in the case of REE-containing samples compared to that of undoped TiO_2. It is worth noting that such intensity growth was stronger for co-doped powders.

For the prepared REE-doped TiO_2 materials, a significant absorption was also registered in the NIR spectral range (800–2000 nm) with distinctive sharp spectral features arising from 4*f* to 4*f* electron transitions of individual REE compounds (Runowski et al., 2020) (Fig. S16). Here, for mono-doped TiO_2, three major bands corresponding to $4f \rightarrow 4f$ absorption are observed for Sm^{3+} at 996, 1085, and, 1235 nm, for Tm^{3+} at 981, 1185, and 1211 nm, and for Yb^{3+} at 946, 977, and 1204 nm, whereas Gd^{3+} and Tb^{3+} ions do not have any significant absorption in this region, and their 4*f*-5*d* absorption bands are located below 400 nm. Such differences in the absorption properties of middle and heavy REE

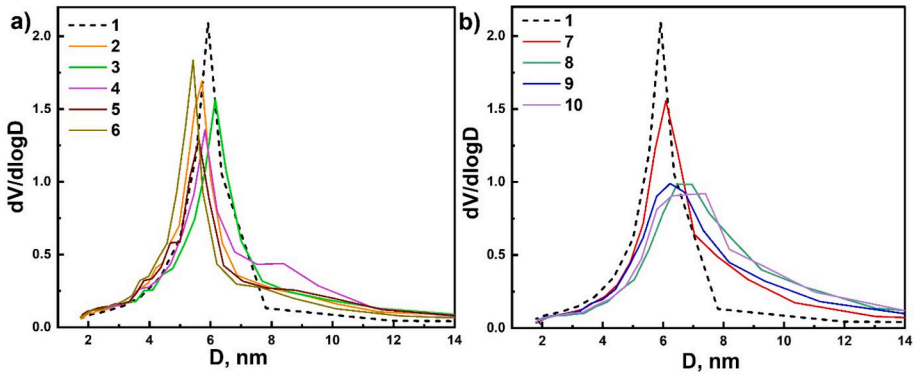


Fig. 5. Pore size distribution curves for the mono- (a) and co-doped (b) materials compared with undoped TiO_2: T (1), T_{Sm} (2), T_{Tb} (3), T_{Gd} (4), T_{Tm} (5), T_{Yb} (6), and T (1), T_{SmTm} (7), T_{SmYb} (8), T_{SmTb} (9), and T_{SmGd} (10).

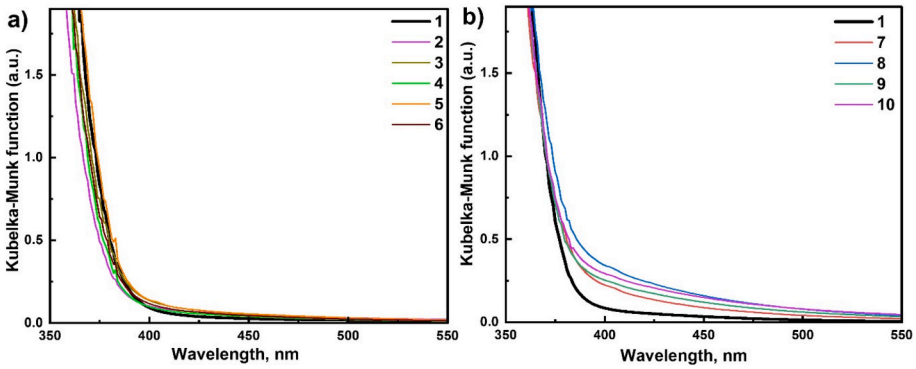


Fig. 6. UV–Vis spectra of the mono-doped (a) and co-doped (b) powders: T (1), T_{Gd} (2), T_{Yb} (3), T_{Tb} (4), T_{Sm} (5), T_{Tm} (6), T_{SmTm} (7), T_{SmYb} (8), T_{SmTb} (9), and T_{SmGd} (10).

can be responsible for their different light-induced activity. The NIR spectra of double-doped samples correspond to a superposition sum of the above-mentioned bands of individual REE ions, which are well overlaid without any changes in their shape or peak position, thus indicating that different dopant ions are rather isolated from each other and are not involved in excitation energy transfer processes.

The calculated energy gap values of REE-doped TiO_2 samples are presented in Table S2. The E_g values of mono-doped samples are negligibly higher (3.18 ± 0.03 eV) compared to the undoped material (3.17 ± 0.03 eV). Such a small difference is within the measurement uncertainty (±0.03 eV) and therefore not considered a statistically resolvable change. It means that the reason for the absorption in the visible light range for the mono-doped materials (Fig. 6, b) is the introduction of localized sub-band gap REE-related energy states rather than actual E_g narrowing of the mono-doped T (Wekalao et al., 2026).

On the contrary, co-doped samples show a small decrease in E_g values (T_{SmTm} – 3.12 ± 0.03 eV, T_{SmTb} and T_{SmGd} – 3.13 ± 0.03 eV, and less for T_{SmYb} – 3.16 ± 0.03 eV), which may relate to the introduction of localized 4*f* impurity energy levels within the forbidden band and oxygen vacancies to balance the charge mismatch of ions. The REE pairs, with their diverse electronic transitions, form a denser continuum of impurity levels and a greater number of oxygen vacancies in the T matrix (Du et al., 2021). The synergistic hybridization of dopants' 4*f* orbitals with O 2*p* and Ti 3*d* states shifts the valence and conduction band edges, effectively reducing the excitation energy and shifting the light absorption into the visible region far more effectively than mono-doping does.

The E_g values of the REE-doped TiO_2 samples are within the range 3.12–3.18 eV (Table S2), indicating that irradiation with wavelength of 266 nm (~4.7 eV) will induce intra-band transitions from valence to conduction band. The backward recombination could occur either non-radiatively, with the production of phonons, or through luminescence centres, with the occurrence of characteristic light emission. PL emission spectra of TiO_2 materials are usually attributed to three kinds of physical origins: recombination of self-trapped excitons, electrons trapped at oxygen vacancies and surface states (defects). The electron recombination with the energy of emitting photons close to the E_g (~400 nm) originates from the recombination of photoexcited electron–hole pairs due to a direct transition from the conduction band (Ghamsari et al., 2016).

If the recombination occurs through REE^{3+} ions, corresponding narrow lines should be detected in the PL emission spectra. Investigation of the emission spectra from REE-doped TiO_2 samples allowed us to estimate whether REE^{3+} ions participate in the photocatalytic process. Fig. 7 (Fig. S17) shows room-temperature light emission spectra of REE-doped TiO_2 samples under the excitation of λ_{exc} = 266 nm.

The emission spectrum from the prepared undoped T exhibits a broad band extending from ~ 350 to 800 nm, with the maximum emission at 445 nm (Fig. 7, a). Such broadband emission most probably involves multiple emission centres, such as inter-band and intra-band transitions, and those related to the defect structure. No characteristic Yb^{3+} emission associated with $2f_{5/2}$-$2f_{7/2}$ electron transition of Yb^{3+} ions was registered from the T_{Yb} sample, which is usually observed in the range from 900 to 1100 nm (Chaika et al., 2022), probably due to the low sensitivity of the NIR detector used. Instead, the T_{Yb} sample exhibits UV–Vis emission spectra similar to those of undoped dual-phase TiO_2, only with higher visible-light emission intensity at ~ 650 nm (Fig. 7, a). This can be due to the formation of colour centres with Yb^{3+} ions (Chaika et al., 2020) or the presence of other defect states within the band gap.

As seen from Fig. 7 c, the mono-doped T_{Tb} samples exhibit characteristic Tb^{3+} emission with the most intense emission bands centred at

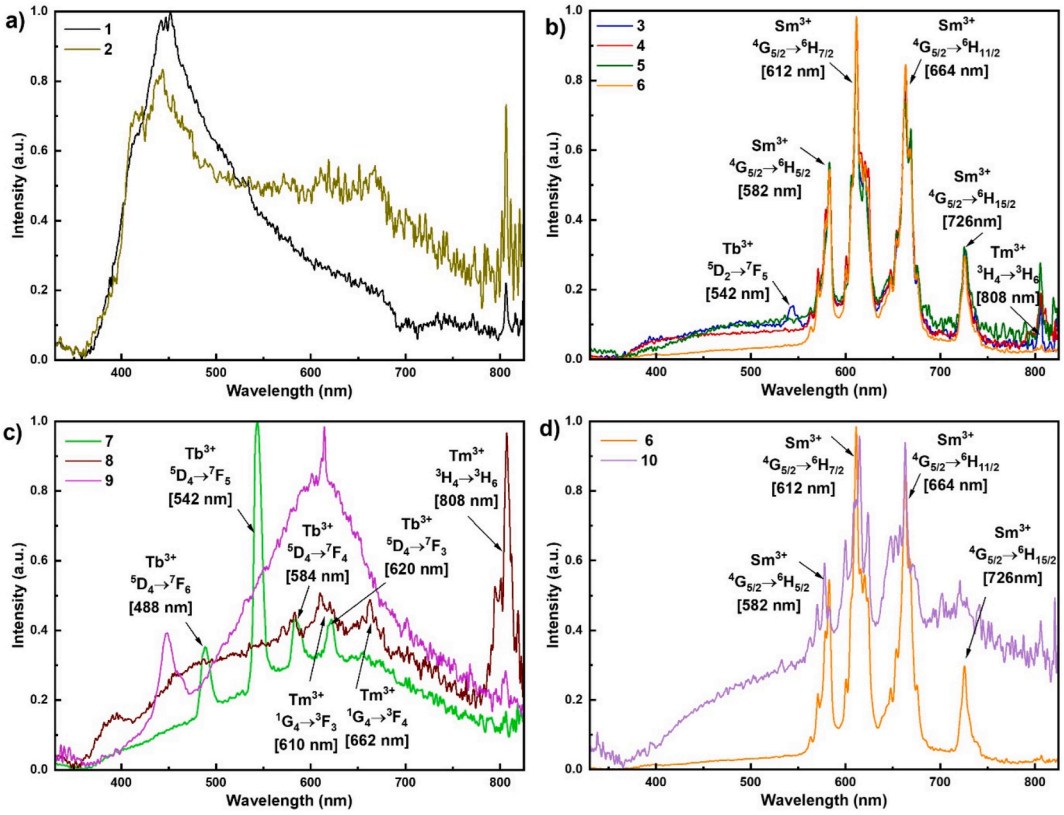


Fig. 7. Normalized luminescence spectra of TiO_2-doped samples under λ_{exc} = 266 nm, T = 295 K, and with measurement step of 1 nm: T (1), T_{Yb} (2), T_{SmTb} (3), T_{SmTm} (4), T_{SmYb} (5), T_{Sm} (6), T_{Tb} (7), T_{Tm} (8), T_{Gd} (9), and T_{SmGd} (10).

488 nm, 542 nm, 584 nm, and 620 nm corresponding, respectively, to $^5D_4 \rightarrow {}^7F_6$, $^5D_4 \rightarrow {}^7F_5$, $^5D_4 \rightarrow {}^7F_4$, and $^5D_4 \rightarrow {}^7F_3$ electronic transitions. Also, in the measured emission spectra of the samples with Tm, emission from Tm^{3+} ions can be recognized with the most intense lines at 610 nm, 662 nm, and 808 nm originating from $^1G_4 \rightarrow {}^3F_3$, $^1G_4 \rightarrow {}^3F_4$, and $^3H_4 \rightarrow {}^3H_6$, transitions, respectively (Fig. 7, c). Investigation of the luminescence spectra from mono-doped T_{Gd} sample reveals a broad emission band centred at ~ 600 nm and a relatively narrow, weaker emission at 448 nm. It is known that when excited with UV light, Gd^{3+} ions commonly show intrinsic emission around 330 nm, and therefore, it cannot be registered in our measurements. Thus, the observed broadband emission ranging from 450 to 800 nm probably indicates an effective energy transfer from Gd^{3+} to Ti^{4+} ions. TiO_2 samples doped with Sm^{3+} (Fig. 7, b) clearly exhibit the emission lines originating from Sm^{3+} ions, showing the strongest lines at 582, 612, 664, and 726 nm, which were assigned to the electronic transitions $^4G_{5/2} \rightarrow {}^6H_{5/2}$, $^4G_{5/2} \rightarrow {}^6H_{7/2}$, $^4G_{5/2} \rightarrow {}^6H_{11/2}$, and $^4G_{5/2} \rightarrow {}^6H_{15/2}$, respectively. Detailed information regarding REE^{3+} energy level structure can be found in the literature (Wang et al., 2018). The presence of REE^{3+} luminescence strongly suggests their involvement in the electron recombination processes, and therefore in the photocatalytic activity of REE-doped TiO_2.

PL emissions from double-doped TiO_2 materials, T_{SmTb}, T_{SmYb}, and T_{SmTm} are all dominated by Sm^{3+} emission bands with no difference in their position compared to mono-doped T_{Sm} (Fig. 7, b, d). In addition, the T_{SmGd} sample exhibits slightly shifted Sm^{3+} emission lines centred at 578, 615, 663, and 720 nm superimposed over a broad visible emission ranging from 450 to 800 nm due to energy transfer between Gd^{3+} to Ti^{4+} ions.

It should be noted that in crystals, the REE^{3+} emission levels are split into Stark components represented in PL spectra as several narrow lines of the same energy level transition (Szysiak et al., 2025). Within the accuracy limit of our measurements, individual Sm^{3+} Stark components retain their positions in the PL spectra of all double-doped TiO_2 compared to mono-doped T_{Sm} material. However, the fine structure of Sm^{3+} emission is most pronounced for the T_{SmGd} sample (Fig. 7, d) compared to the other samples with different intensity distributions. This suggests changes in the nearest crystal field environment surrounding Sm^{3+} ions in the T_{SmGd} as compared to other REE-doped materials under study (Gluchowski and Chaika, 2024). Together with the distinct luminescence properties of Gd-doped TiO_2, this might indicate that double-doping of TiO_2 with Sm^{3+} and Gd^{3+} influences the microstructure of the TiO_2 host.

3.5. Photocatalytic measurements

The degradation reaction of the commercial drug DCF under UV and visible light was chosen for testing the photocatalytic activity of the REE-doped powders. Prior to irradiation, the photocatalyst-DCF systems were kept in the dark for two hours to achieve adsorption equilibrium.

Control experiments showed that, in the absence of a photocatalyst, DCF did not undergo photolysis under either UV or visible light. Photodegradation of DCF began only in the presence of the prepared REE-doped powders. XRD analysis confirmed that the powder structure remains unchanged after photocatalysis, indicating good structural stability (Fig. S18). The photocatalytic performance and adsorption capacity of the synthesized materials under UV and visible light are shown in Fig. 8 and Fig. S19.

Compared to undoped TiO_2, REE-doped samples show a higher adsorption capacity for DCF. Co-doped samples, despite their reduced surface area, display better adsorption of DCF compared to mono-doped materials. The main reason can be the presence of larger pores in their mesoporous structure, which provide better access for DCF molecules to the pore surface (Table 1, Fig. 5). This can be confirmed by the T_{SmTm} sample with a pore size distribution close to that of T_{Sm} and T_{Tm} (Fig. 5) and shows a similar adsorption capacity for DCF (Fig. 8). The percentage of DCF removal in the presence of undoped TiO_2 was 43.6 % and 1.5 %

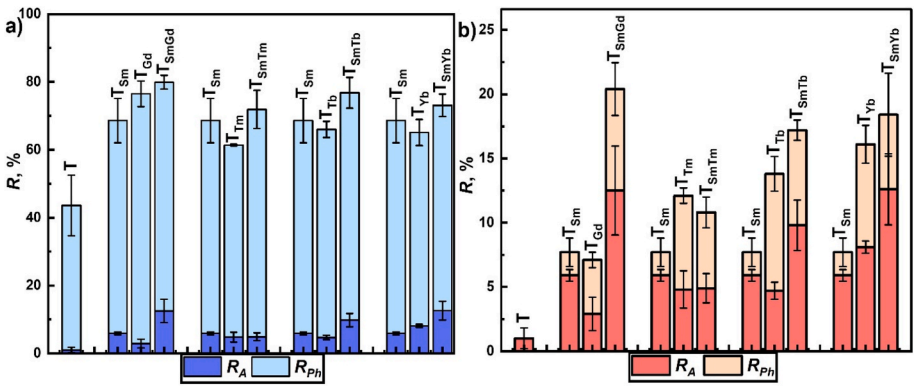


Fig. 8. Photocatalytic (R_{Ph}) and adsorption (R_A) performance of the prepared materials in the DCF photodegradation reaction under UV (a) and visible light (b).

under UV and visible light, respectively. The use of REE-doped powders increased this value up to 76.5 % (T_{Gd}) and 79.9 % (T_{SmGd}) under UV (Fig. 8, a) and to 16.1 % (T_{Tb}) and 20.4 % (T_{SmGd}) under visible light (Fig. 8, b) for mono- and co-doped materials, respectively.

The rate of photocatalytic DCF degradation is also higher for REE-doped samples as compared with undoped TiO_2 (Fig. 9, Table S3).

The pseudo-first-order kinetic model provided an excellent fit to the DCF photodegradation data under UV irradiation for all samples (R^2 = 0.993–0.999, Table S3), confirming that this model adequately describes the reaction kinetics under UV light across the entire series. Mono-doping improved the DCF degradation rate by 1.5–2.6 times under UV light compared with undoped TiO_2, suggesting effective sensitization of TiO_2 by REE ions (Fig. 9, a). The most effective among the mono-doped samples was T_{Gd}. Such significant effectiveness of this dopant may be related to the crucial role of its highly stable half-filled 4f electronic configuration (Xu et al., 2002) that can be reduced by the trapped electrons during the photocatalytic process. Consequently, electrons can be more easily transferred, thereby restoring the electronic structure and accelerating the photocatalytic process. Indeed, Gd-doping significantly increases TiO_2 photocatalytic performance (Xu et al., 2002; Khalyavka et al., 2025). Co-doping further slightly increases the rate of the reaction under UV light in the case of T_{SmTb} and T_{SmTm} (by1.4–1.5 times), while having no effect on T_{SmGd} and T_{SmYb} materials compared with mono-doping (Fig. 9, a).

Under visible-light irradiation, the fit quality of the pseudo-first-order kinetic model was generally good for the REE-doped samples (R^2 = 0.908–0.998), but noticeably lower for undoped TiO_2 (R^2 = 0.477) and, to a lesser extent, for T_{Gd} (R^2 = 0.881). Such a low R^2 for T does not indicate a breakdown of the pseudo-first-order model itself but rather reflects the very low DCF conversion exhibited by this sample under visible light (Fig. 9, a). Indeed, the undoped TiO_2 exhibited negligible activity under visible light > 450 nm, which may relate to the possibility that DCF molecules adsorbed on the TiO_2 surface harvest visible light. Both the mono- and double-REE-doped TiO_2 showed higher photocatalytic activity under visible light (>450 nm) (Fig. 9, b), with $R^2 \geq 0.90$, supporting the validity of the pseudo-first-order model for the visible-light-active compositions. The rate of the reaction increased by 20 (T_{Tm}) and 25 (T_{SmTb}) times, compared with undoped TiO_2 indicating the role of REE as dopants. The most active appeared to be heavy REE-doped T_{Tm}, T_{Tb} and T_{Yb} materials due to their enhanced Vis-NIR absorption. Addition of Sm^{3+} with its effective absorption in the vis-NIR spectral range promotes the photocatalytic activity of T_{SmGd} and T_{SmTb} (nearly twofold). At the same time, the visible-light-induced photoactivity of T_{SmTm} and T_{SmTb} remained unchanged or even decreased. The registered higher UV photocatalytic activity of double-doped T_{SmGd} and T_{SmTb} samples as compared to the mono-doped ones might be due to the favourable relative alignment of the electron energy levels of Sm^{3+}/Gd^{3+} and Sm^{3+}/Tb^{3+} within the band gap of the TiO_2 as well as their possible effect on the microstructure of the TiO_2 host.

A plausible schematic mechanism of DCF photocatalytic degradation, illustrated by the example of the T_{Gd} and T_{SmGd} samples is proposed in Fig. 10.

Under UV excitation, the electron-holes pairs ($e^- - h^+$) are generated within the anatase and brookite phases of the two-phase TiO_2. The excited electrons migrate from brookite to anatase due to their different potentials, promoting interfacial charge separation (Nachit et al., 2022). Under visible light, these processes are not possible for the undoped sample, whereas doped materials can harvest visible light due to the presence of additional REE energy levels and their direct excitation (Fig. 6). REE ions create new charge-transfer pathways, prolonging carrier lifetime and suppressing electron-hole recombination by trapping photogenerated electrons at the impurity energy levels formed by REE *f*-orbitals (Choi et al., 1994). In the co-doped materials, dopants act together to create additional pathways, thereby boosting photocatalyst' performance.

The photogenerated trapped and separated charge carriers migrate

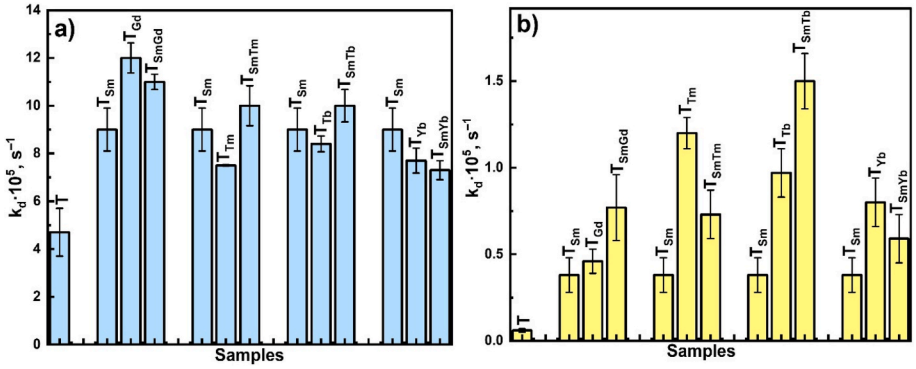


Fig. 9. Kinetic rate constants for DCF photodegradation in the presence of the prepared materials under UV (a) and visible (b) light.

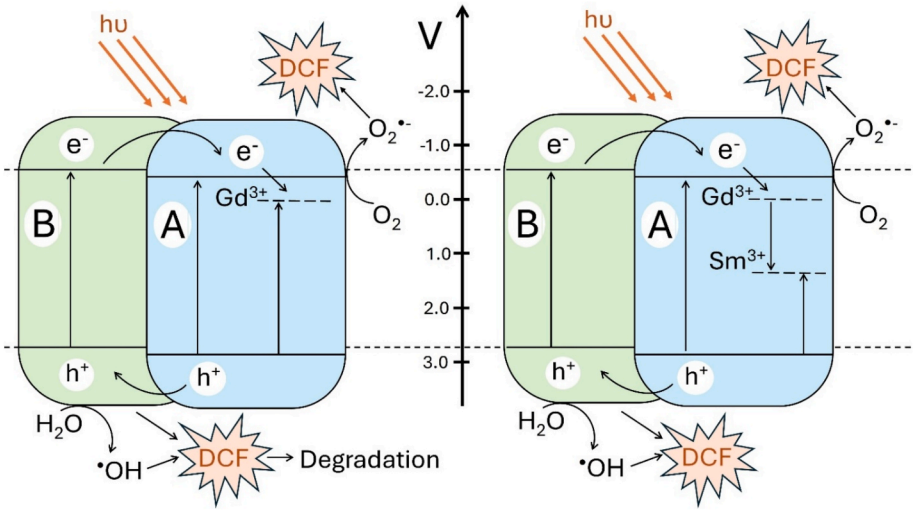


Fig. 10. Schematic diagram illustrating the photocatalytic degradation of DCF in the presence of T_{Gd} and T_{SmGd}.

to the surface of the catalyst, where they participate in surface redox reactions. Holes oxidize adsorbed water and hydroxyl groups to generate hydroxyl radicals ($^{\bullet}OH$) (+2.32 V (Mandari et al., 2018) or directly oxidize adsorbed DCF, while electrons reduce adsorbed O_2 to superoxide radical anions ($O_2^{\bullet-}$) (−0.16 V (Mandari et al., 2018). Given the estimated oxidation potential of DCF (~+1.0 V (He et al., 2013), both direct hole and indirect radicals' oxidation of DCF are thermodynamically favourable. These reactive species attacks DCF molecules, leading to degradation (Yilmaz and Ciltas, 2014).

During UV and visible-light irradiation of the DCF solution in the absence of a photocatalyst, no significant changes in the absorption spectra were observed (Fig. S20, a, b). The irradiation of DCF in the presence of a photocatalyst under visible light led to a decrease in the intensity of the characteristic DCF-absorption band at 275 nm (Fig. S20, c), which is assigned to the n-π* transition of the diphenylamine NH-group. It should be noted that benzyl alcohol, one of the excipients present in the commercial DCF drug used in this study, absorbs in a similar spectral region ($\lambda_{max} \approx 259$ nm). However, due to its low concentration relative to DCF in the working solution, a significant contribution of this excipient to the observed spectra has not been determined.

Under UV light, the DCF absorption band at 275 nm exhibited a shift towards shorter wavelengths (down to ~ 256–259 nm after 3 h), accompanied by a decrease in overall absorbance intensity and an apparent isosbestic point at ~ 310–315 nm (Fig. S20, d). This behaviour is consistent with the gradual formation of aromatic degradation intermediates possessing reduced π-conjugation relative to the diphenylamine chromophore of DCF, in agreement with reports of similar spectral shifts observed for other pharmaceutical compounds (Daescu et al., 2020). The detailed study of these transformations was beyond the scope of the present work and represents an important direction for future research.

Taking into account that the commercial DCF drug contains a low concentration of excipients, it is worth noting that they can potentially affect the adsorption and photocatalytic performance of the studied samples. The adsorption of DCF molecules occurs due to the carboxylate groups that are chemically bound to the TiO_2 surface (Lara-Pérez et al., 2020). Excipients such as mannitol (Al-Enizi et al., 2020) and propylene glycol (Chen et al., 2014) adsorb weakly on the TiO_2 surface via hydrogen bonding. They may act either as hole (h^+) scavengers, suppressing electron-hole recombination and enhancing photocatalytic activity or as hydroxyl radical scavengers, competing with DCF and inhibiting photocatalysis (Denisov et al., 2019). Dissociated into inorganic anions, sodium metabisulfite, may also compete with DCF for radicals and holes and potentially inhibit the photocatalytic process (Schneider et al., 2020). Since the working solution had a pH close to the isoelectric point of TiO_2, where its surface charge was nearly neutral, electrostatic adsorption of these anions is weak. Benzyl alcohol interacts directly with surface hydroxyl groups on the TiO_2, releasing water and forming stable surface-bound titanium alkoxide species (Kobayashi and Higashimoto, 2015). Being a hole scavenger (Zhao et al., 2022), it may inhibit the photocatalytic process. Taken together, these results suggest that the rate of the studied photocatalytic reaction could be higher for a pure DCF solution or a different amount of catalyst (Lee et al., 2021; Lara-Pérez et al., 2020; Kapinus et al., 2009) (Table S4).

4. Conclusions

Mono- and co-doping of two-phase anatase-brookite TiO_2 with rare earth elements (REE: Sm, Gd, Tb, Tm, Yb), prepared by the same method, differently affect the morphological, structural, textural, optical, and photocatalytic properties. XRD, Raman, FTIR, and HR-TEM confirmed that the dual-phase anatase-brookite structure was preserved after both mono- and co-doping, with REE ions accommodated primarily at interfacial and near-surface sites rather than substituting into the TiO_2 lattice, owing to the large ionic radius mismatch between REE^{3+} and Ti^{4+}. REE-doping induced lattice distortion and contraction of the unit cell of TiO_2. Compared to pristine TiO_2, the mono-doped powders exhibited a larger surface area despite a slight decrease in crystallite size. Conversely, co-doped materials had a lower surface area yet possessed a higher pore volume and a larger mesopore size. UV–Vis diffuse reflectance spectroscopy showed that REE-doping enhanced visible-light absorption, with the effect being stronger for co-doped materials due to the synergistic hybridization of dopants' 4*f* orbitals with O 2*p* and Ti 3*d* states. Photoluminescence spectroscopy confirmed that modification with REE facilitated charge carrier migration and effectively inhibited electron-hole recombination. Consequently, the synergistic effects of the anatase/brookite junctions, the developed porous structure, and presence of dopant significantly enhanced both the adsorption capacity and the photocatalytic activity under UV and visible light. In photodegradation tests of diclofenac (DCF), the most active catalyst appeared to be TiO_2 co-doped with Sm and Gd, which showed drug removal rates of 20.4 % and 79.9 % under visible (>450 nm) and ultraviolet light, respectively. Furthermore, the photocatalytic rate constants for DCF degradation by the doped materials were 1.5–2.6 times higher than those of pristine TiO_2. The results of this work will be useful for the optimal design and fabrication of sunlight-active photocatalysts for pharmaceutical waste decomposition in aquatic environment.

CRediT authorship contribution statement

Tetiana Khalyavka: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Regina Abdurrahmanoglu:** Writing – review & editing, Investigation, Formal analysis. **Nataliya Shcherban:** Writing – review & editing, Investigation. **Mykhailo Chaika:** Writing – original draft, Investigation, Formal analysis. **Tetiana Gavrilko:** Writing – original draft, Investigation, Formal analysis. **Oscar Cabezuelo:** Writing – review & editing. **Oleg Tkachenko:** Writing – review & editing, Investigation, Formal analysis. **Elif Coşkun:** Writing – review & editing, Investigation. **Pavel Yaremov:** Writing – review & editing, Investigation. **Tetyana M. Budnyak:** Writing – review & editing, Resources. **Jean-Claude Grivel:** Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Funding for this research was provided by the SARU fellowship program for Ukrainian researchers at risk and the Technical University of Denmark (DTU). The studies were also partially supported by the EduSustRec4Ukraine (Baltic Sea Neighbourhood Programme, BSNP 01257-2023) research project, funded by the Swedish Institute and the Wallenberg Initiative Materials Science for Sustainability (WISE), funded by the Knut and Alice Wallenberg Foundation, and by Smart Exploration Research Center funded by the Swedish Foundation for Strategic Research (SSF), agreement no. CMM22-0003. T.G. is deeply grateful to the Polish Academy of Sciences for the fellowship provided under the 'Long-Term Program of Support for Ukrainian Researchers', organized in collaboration with the U.S. National Academy of Sciences.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ces.2026.125093.

Data availability

Data will be made available on request.

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